

JOSIAH DANIEL D. CATABAY

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SKILLS

Languages: Python, JavaScript, PHP, Java, SQL

Framework & Libraries: React, Vite, Tailwind CSS, Bootstrap

ML/NLP: Random Forest, Whisper, BART, Transformers, PyTorch, LLM Integration

Backend: REST APIs, Model Inference APIs, JWT Authentication

Data: MySQL, Firebase, Supabase

Blockchain: Solidity, Ethereum

SCHOOL PROJECTS

CropForecast— AI Driven Agricultural Decision Support System

Full Stack Engineer

- Achieved $R^2 = 0.9929$, **RMSE = 2,221.70**, and **MAE = 544.44** on a **32,928**-row holdout evaluation for model v1.2, demonstrating strong predictive accuracy on unseen production data.
 - Architected a shared **Flask backend** powering **React web and Android** clients through **91 REST endpoints** spanning authentication, farms, recommendations, crop analytics, admin operations, knowledge workflows, and ML lifecycle services.
 - Developed a role-based **React SPA** with 37 route definitions and protected navigation across Farm Owner, Analyst, Admin, Superadmin, and ML Engineer workspaces, improving workflow isolation and usability.
 - Engineered a 165,336-row agricultural forecasting dataset spanning 2010-2025, 53 provinces, and 89 crops, then refreshed it to 167,699 rows for the 2025Q4 release through a repeatable quarterly update pipeline.
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DagupanGovLedger — Blockchain Transparency Platform + LLM Assistant

Lead Engineer

- Built a **blockchain-backed public spending platform** with immutable transaction records using **Ethereum smart contracts**.
 - Designed smart-contract workflows storing **on-chain hashes + off-chain metadata** for auditability.
 - Integrated an **LLM assistant** to explain budgets, audits, and spending data in natural language.
 - Owned backend APIs for authentication, role-based access, audits, and analytics.
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PyVidNote — NLP Video Transcription & Summarization

ML Engineer

- Built a **Python mobile app** converting video → text → summary via **Whisper (ASR)** and **BART (abstractive summarization)**.
 - Implemented an end-to-end NLP inference pipeline optimized for usability and fast review.
 - Delivered a mobile-friendly UI enabling file selection and instant content extraction.
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EXPERIENCE

Web Developer (OJT) — PHINMA University of Pangasinan | 2023–2024

- Delivered a production-ready web system recognized with **Best in Business Venture** award.
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EDUCATION

B.S. in Information Technology — PHINMA University of Pangasinan (2024–Present)